

Barton Springs Salamanders

Jamhad Byers & Nick Gonzalez

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0.1 Barton Springs Salamanders

The Barton Springs Salamanders DO and Flow data-set from the City of Austin provides measurements of dissolved oxygen (DO) and spring flow rates at Barton Springs, an important habitat for the endangered Barton Springs Salamander (*Eurycea sosorum*). The data-set includes time-series records of DO concentrations (in mg/L) and spring flow (in cubic feet per second) collected from various sites within the Barton Springs complex. These environmental factors are critical indicators of water quality and habitat health for the salamander population. By analyzing trends in DO levels and flow rates, researchers and conservationists can assess the impact of environmental changes, groundwater conditions, and human activity on the ecosystem.

This dataset is interesting due to its relevance in the broader conversation about climate change and the methods used to keep track of the nuanced and widespread effects. For our purposes here, the data will be used to asses dissolved oxygen levels, salamander counts, flow severity, and sample areas of four sites in the Austin area. Each row represents a measurement on a particular day at a particular site.

0.1.1 Original Variables

`site_name` is, as the name suggest, the name of the site for that particular instance of data collection.

`sample_date` is also as it's name suggest, in that it list the date that sample was taken.

`parameter result` & `unit` are the measurement, numeric value of that measurement, and the units of that measurement respectively

0.1.2 Created Variables

DO is created from the Dissolved Oxygen value in PARMETER, and contains the numeric values of the measurement.

Flowcode is created from the FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY) value in PARMETER, and is categorical representation by the Texas Commission on Environmental Quality (TCEQ) of the type of flow observed. The numbers correspond to their listed type.

salamanders is created from the NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED value in 'PARMETER', and is a count of how many salamanders were seen during that particular measurement.

Finally, area is created from the AREA SAMPLED value in PARMETER, and is a measure of the total square footage that was survived for a particular site on a particular day.

The data is provided by the Barton Springs Salamanders DO and Flow data-set from the City of Austin and is a "view" of the Water Quality Sampling Data dataset - in that the Parameters and sites and projects are limited to those related to this particular subject (Barton Springs Salamanders, Flow and DO).

Before cleaning, the set contains 50,068 rows and 22 columns.

0.2 Methods

This analysis investigates how dissolved oxygen (DO) levels vary across different Barton Springs sites, and whether that variation is confounded by differences in survey effort (area sampled) or salamander detection rates across sites. An initial question regarding whether filter fraction (dissolved vs. total) affects DO readings was set aside, as filter type was inconsistently recorded across site visits and would have required a separate data structure to address properly. This is a natural extension for future work.

0.2.1 Loading Packages

```
# Upload / install packages
library(tidyverse)
```

```
## Warning: package 'lubridate' was built under R version 4.4.1
library(plotROC)
```

0.2.2 Importing the Data

```
water <- read_csv("Barton_Springs_Salamanders_DO_and_Flow_20250305.csv")
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
# data preview
head(water)
```

```
## # A tibble: 6 x 22
##   WATERSHED  SAMPLE_DATE SITE_NAME LON_DD_WGS84 LAT_DD_WGS84 SITE_TYPE MEDIUM
##   <chr>      <chr>      <chr>      <dbl>      <dbl> <chr>      <chr>
```

```
## 1 Barton Creek 02/25/2025 ~ Barton S~ -97.8 30.3 Spring Groun~
## 2 Barton Creek 01/08/2025 ~ Barton S~ -97.8 30.3 Spring Groun~
## 3 Barton Creek 12/11/2024 ~ Old Mill~ -97.8 30.3 Spring Groun~
## 4 Barton Creek 12/11/2024 ~ Barton S~ -97.8 30.3 Spring Groun~
## 5 Barton Creek 12/11/2024 ~ Eliza Sp~ -97.8 30.3 Spring Groun~
## 6 Barton Creek 11/21/2024 ~ Barton S~ -97.8 30.3 Spring Habit~
## # i 15 more variables: PARAM_TYPE <chr>, PARAMETER <chr>, QUALIFIER <lg1>,
## # RESULT <dbl>, UNIT <chr>, FILTER <chr>, SAMPLE_ID <chr>,
## # SAMPLE_SITE_NO <dbl>, METHOD <chr>, QC_FLAG <chr>, PROJECT <chr>,
## # DATA_REF_NO <dbl>, SAMPLE_REF_NO <dbl>, TIME_NULL <chr>, QC_TYPE <lg1>
```

```
summary(water)
```

```
## WATERSHED      SAMPLE_DATE      SITE_NAME      LON_DD_WGS84
## Length:50068   Length:50068      Length:50068   Min.    :-97.77
## Class :character Class :character Class :character 1st Qu.: -97.77
## LAT_DD_WGS84   SITE_TYPE      MEDIUM      PARAM_TYPE
## Min.    :30.26 Length:50068      Length:50068   Length:50068
## 1st Qu.:30.26 Class :character Class :character Class :character
## PARAMETER      QUALIFIER      RESULT      UNIT
## Length:50068   Mode:logical   Min.    : -5.50 Length:50068
## Class :character NA's:50068     1st Qu.:  0.00 Class :character
## FILTER      SAMPLE_ID      SAMPLE_SITE_NO METHOD
## Length:50068 Length:50068     Min.    : 35.0 Length:50068
## Class :character Class :character 1st Qu.: 183.0 Class :character
## QC_FLAG      PROJECT      DATA_REF_NO  SAMPLE_REF_NO
## Length:50068 Length:50068     Min.    : 91172 Min.    : 9986
## Class :character Class :character 1st Qu.:1692903 1st Qu.:305031
## TIME_NULL    QC_TYPE
## Length:50068 Mode:logical
## Class :character NA's:50068
## [ reached getOption("max.print") -- omitted 5 rows ]
```

0.2.3 Choosing 4 Sites for Analysis

```
# site options
unique(water$SITE_NAME)
```

```
## [1] "Barton Spring"
## [2] "Old Mill (Sunken Gardens) Spring"
## [3] "Eliza Spring"
## [4] "Old Mill Stream (outflow from Sunken Gardens Pool)"
## [5] "Upper Barton Spring"
## [6] "High Barton Spring"
```

Our analysis will focus on the Barton Spring, Eliza Spring, Old Mill (Sunken Gardens) Spring, and Upper Barton Spring sites.

0.2.4 Understanding Filter Fraction

```
unique(water$FILTER)
```

```
## [1] "Dissolved" "Total"      "N/A"
```

“Total” samples includes everything, from dissolved stuff tp particles floating in the sample (suspended solids). “Dissolved” samples only include the dissolved substances, the sample is

passed through a filter first, which removes solids. “N/A” is currently unknown.

0.2.5 Choosing Parameters to Review

The paramters variable is a representation of what measurement was taken for that row. Here, we display all available measurements for analysis in the dataset

```
# list of parameters
unique(water$PARAMETER)

## [1] "DISSOLVED OXYGEN"
## [2] "TOTAL TIME SPENT"
## [3] "NUMBER OF BARTON SPRINGS SALAMANDERS PHOTOGRAPHED"
## [4] "NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED"
## [5] "FLOW"
## [6] "AUSTIN BLIND SLMNDR (1-2IN.)"
## [7] "AUSTIN BLIND SLMNDR (>=2IN.)"
## [8] "BARTON SPRINGS SLMNDR (TOTAL 1-2IN.)"
## [9] "BARTON SPRINGS SLMNDR (TOTAL >=2IN.)"
## [10] "BARTON SPRINGS SLMNDR (TOTAL <1IN.)"
## [11] "AUSTIN BLIND SLMNDR (<1IN.) (JUVENILE)"
## [12] "FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY)"
## [13] "SUM OF BSS SALAMANDERS PER TRANSECT"
## [14] "SUM OF ABS SALAMANDERS PER TRANSECT"
## [15] "BASEFLOW"
## [16] "AUSTIN BLIND SALMNR (ADULT) > 1 INCH"
## [17] "BARTON SPGS SALMNR (ADULT) > 1 INCH"
## [18] "SALAMANDER (UNKNOWN LENGTH)"
## [19] "SALAMANDERS PHOTGRAPHED >2 INCHES"
## [20] "SALAMANDERS PHOTGRAPHED <1 INCH"
## [21] "SALAMANDERS PHOTGRAPHED 1-2 INCHES"
## [22] "SALAMANDERS NOT PHOTGRAPHED >2 INCHES"
## [23] "SALAMANDERS NOT PHOTGRAPHED <1 INCH"
## [24] "SALAMANDERS NOT PHOTGRAPHED 1-2 INCHES"
## [25] "AUSTIN BLIND SALAMANDER (UNKNOWN LENGTH)"
## [26] "BARTON SPRINGS SLMNDR (1-2IN.) IN COBBLE/BOULDER"
## [27] "BARTON SPRINGS SLMNDR (>=2IN.) IN COBBLE/BOULDER"
## [28] "BARTON SPRINGS SLMNDR (1-2IN.) IN VEGETATION/MOSS"
## [29] "BARTON SPRINGS SLMNDR (>=2IN.) IN VEGETATION/MOSS"
## [30] "AUSTIN BLIND SLMNDR (1-2IN.) ON EXPOSED SUBSTRATE"
## [31] "AUSTIN BLIND SLMNDR (>=2IN.) ON EXPOSED SUBSTRATE"
## [32] "AUSTIN BLIND SLMNDR (1-2IN.) IN LEAF LITTER"
## [33] "AUSTIN BLIND SLMNDR (>=2IN.) IN LEAF LITTER"
## [34] "AUSTIN BLIND SLMNDR (>=2IN.) IN COBBLE/BOULDER"
## [35] "AUSTIN BLIND SLMNDR (1-2IN.) IN COBBLE/BOULDER"
## [36] "AUSTIN BLIND SLMNDR (1- 2IN.) IN VEGETATION/MOSS"
## [37] "AUSTIN BLIND SLMNDR (>=2IN.) IN VEGETATION/MOSS"
## [38] "BARTON SPRINGS SLMNDR (<1IN.) ON EXPOSED SUBSTRATE"
## [39] "BARTON SPRINGS SLMNDR (<1IN.) IN LEAF LITTER"
## [40] "BARTON SPRINGS SLMNDR (<1IN.) IN COBBLE/BOULDER"
## [41] "AUSTIN BLIND SLMNDR (<1IN.) ON EXPOSED SUBSTRATE"
## [42] "AUSTIN BLIND SLMNDR (<1IN.) IN LEAF LITTER"
## [43] "AUSTIN BLIND SLMNDR (<1IN.) IN COBBLE/BOULDER"
## [44] "AUSTIN BLIND SLMNDR (<1IN.) IN VEGETATION/MOSS"
## [45] "BARTON SPRINGS SLMNDR (<1IN.) IN VEGETATION/MOSS"
```

```
## [46] "BARTON SPRINGS SLMNDR (>=2IN.) ON EXPOSED SUBSTRATE"
## [47] "BARTON SPRINGS SLMNDR (1-2IN.) ON EXPOSED SUBSTRATE"
## [48] "BARTON SPRINGS SLMNDR (1-2IN.) IN LEAF LITTER"
## [49] "BARTON SPRINGS SLMNDR (>=2IN.) IN LEAF LITTER"
## [50] "AREA SAMPLED"
## [ reached getOption("max.print") -- omitted 12 entries ]
```

The definitions of these were provided by Mr. Robert Clayton from the City of Austin Watershed Protection Department. For this analysis, the measurements of interest include DISSOLVED OXYGEN, FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY), NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED, and AREA SAMPLED.

0.2.6 Associated Units List

The units of interest all correspond to the measurements of interest, so we removed all unnecessary unit types from the dataset for analysis

```
unique(water$UNIT)
```

```
## [1] "MG/L"
## [2] "Minutes"
## [3] "Count"
## [4] "Cubic Feet Per Second"
## [5] "None"
## [6] "1-2: (Yes; No)"
## [7] "Feet per second"
## [8] "(%) Percent"
## [9] "Sq Ft"
## [10] "Inches"
## [11] "Feet"
## [12] "MM"
## [13] "gal/min"
## [14] "Relative Abundance (1): <20 (2): 21-50 (3): >50"
```

The units of interest are “MG/L”, “Sq Ft”, “None”, and “Count”. “None” is there for the flowcode category.

0.2.7 Data Cleaning

The raw dataset stores each measurement as its own row, with the specific measurement type recorded in `PARAMETER` and its value in `RESULT`. This makes metrics from a single site visit appear as different observations (rows). To analyze relationships between these variables, the data needs to be reshaped so that each row represents a single site visit, with each measurement type as its own column.

Several complications arose in this reshaping process, worth noting explicitly:

- 1.) Some site visits included multiple sub-samples (i.e measurements taken at different depths or along different survey transects on the same day). Rather than treating these as separate observations (which would fragment a single visit across multiple rows) or arbitrarily keeping only one, replicate values were averaged together to produce a single representative value per site per day.
- 2.) The original sample date included exact time of day, and different measurements taken during the same visit were sometimes logged at slightly different timestamps. Grouping by the full timestamp caused measurements from the same visit to be split into separate rows. This was resolved by grouping on the calendar date alone, which correctly reunites same-day measurements.

3.) An earlier version of the cleaning step used to filter for site names used `==` to filter to the four sites of interest, which unintentionally checks values by position rather than membership, silently dropping some valid rows. This was corrected using `%in%`, which properly matches each row against the full list of target sites regardless of order.

4.) Some Dissolved Oxygen readings were recorded in “% Saturation” rather than mg/L, alongside a separate, correctly-scaled mg/L reading for the same site visit. Since these two measurements represent fundamentally different scales, averaging them together during pivoting produced physically implausible DO values (e.g., 60+ mg/L). This was resolved by *excluding percent-saturation readings* before reshaping, ensuring *only mg/L values were averaged into the DO column*.

After reviewing the original dataset and what is important for this analysis, the following is the cleaning done to achieve the desired new dataset to work with.

```
# cleaned data, all managed in one spot and edited throughout the analysis
water_cleaned <- water |>
  # exclude percent-saturation readings
  filter(!(PARAMETER == "DISSOLVED OXYGEN" & UNIT == "(%) Percent")) |>
  # uniform date format
  mutate(sample_day = as_date(mdy_hms(SAMPLE_DATE))) |>
  pivot_wider(
    id_cols = c(sample_day, SITE_NAME),
    # the values of the variable `parameter` will become variables
    names_from = PARAMETER,
    # the cell values of `result` will match the corresponding variable
    values_from = RESULT,
    # averages replicate sub-samples
    values_fn = ~ mean(.x, na.rm = TRUE)
  ) |>
  # selecting variables for analysis and changing names
  select(
    sampledate = sample_day,
    site = SITE_NAME,
    DO = `DISSOLVED OXYGEN`,
    Flowcode = "FLOW SEVERITY CODE (1=NONE;2=LOW;3=NORM;4=FLOOD;5=HIGH;6=DRY)",
    salamanders = "NUMBER OF AUSTIN BLIND SALAMANDERS PHOTOGRAPHED",
    area = "AREA SAMPLED",
  ) |>
  # choosing specific sites
  filter(site %in% c("Barton Spring",
                    "Eliza Spring",
                    "Old Mill (Sunken Gardens) Spring",
                    "Upper Barton Spring"))

head(water_cleaned)
```

```
## # A tibble: 6 x 6
##   sampledate site          DO Flowcode salamanders area
##   <date>      <chr>      <dbl>   <dbl>      <dbl> <dbl>
## 1 2025-02-25 Barton Spring 4.61      NA         NA      NA
## 2 2025-01-08 Barton Spring 4.29      NA         NA      NA
## 3 2024-12-11 Old Mill (Sunken Gardens) Spring 2.63      NA         NA      NA
## 4 2024-12-11 Barton Spring 4.18      NA         NA      NA
## 5 2024-12-11 Eliza Spring 4.36      NA         NA      NA
```

```
## 6 2024-11-21 Barton Spring          NA          NA          0          NA
```

After cleaning, the dataset contains site-visit-level observations for the four springs of interest, with dissolved oxygen, flow severity, salamander counts, and survey area as separate columns. It's worth noting that DO is measured far more frequently and consistently than salamander counts or area surveys, which appear to follow a more occasional or opportunistic sampling schedule. This is reflected in the higher proportion of missing values for those two variables. This is a limitation to keep in mind when interpreting any results involving salamander presence or survey area.

0.3 Results

0.3.0.1 Summary Stats Before comparing DO across sites, it's worth first checking the overall distribution of DO values and how completely it was recorded at each site. This establishes a baseline understanding of the variable itself, and confirms that missing data isn't concentrated at any one site in a way that could bias later comparisons.

```
summary(water_cleaned$DO)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##    0.000   5.140   5.970   5.945   6.670   21.470     893
```

```
water_cleaned |>
  group_by(site) |>
  summarise(n = n(),
            n_missing_DO = sum(is.na(DO))
  )
```

```
## # A tibble: 4 x 3
##   site                                n n_missing_DO
##   <chr>                          <int>      <int>
## 1 Barton Spring                  964         343
## 2 Eliza Spring                   687         227
## 3 Old Mill (Sunken Gardens) Spring 542         222
## 4 Upper Barton Spring            319         101
```

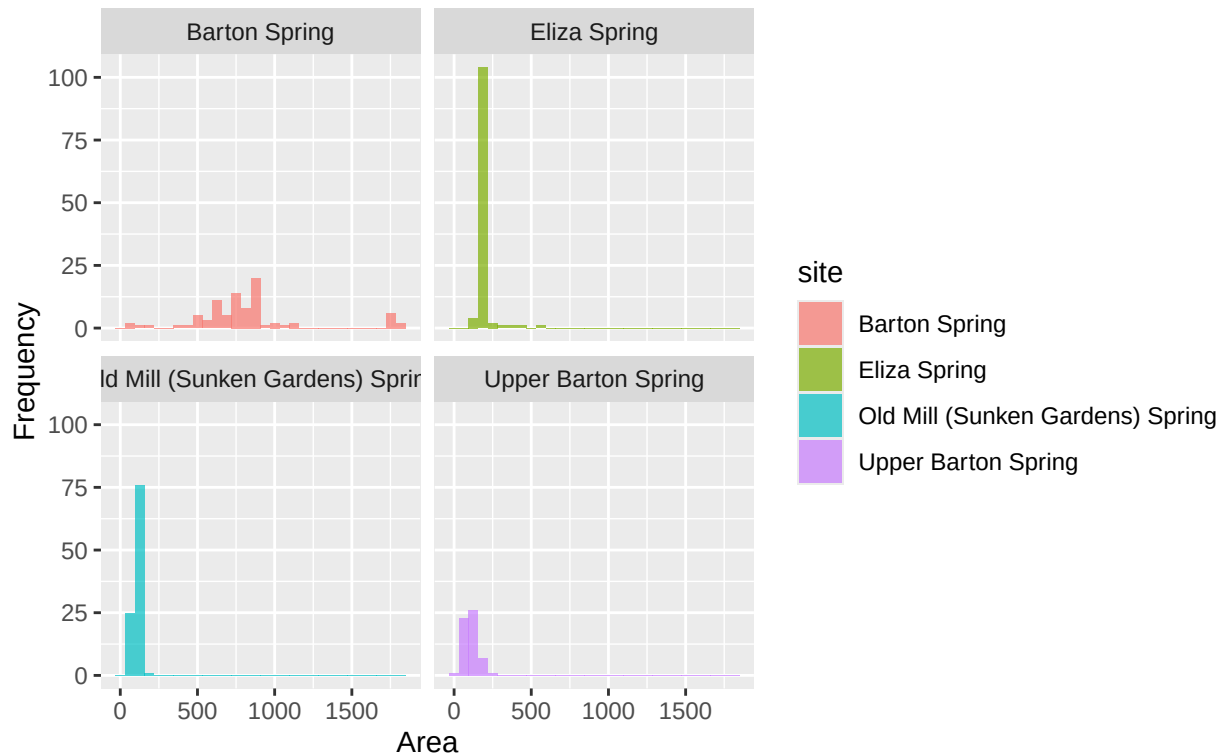
Missing DO readings are broadly similar in proportion across all four sites (32-41%), suggesting no single site's DO comparison is disproportionately affected by data availability.

Before directly addressing how dissolved oxygen (DO) levels vary across different spring sites, and what is their relationship with seasonal changes is, it is important to know some things about the sites themselves. First, we looked at what the variation in areas sampled for each site looked like

```
water_cleaned |>
  ggplot() +
  geom_histogram(aes(x = area, fill = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Variation in Survey Area by Site",
       x = "Area",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2152 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

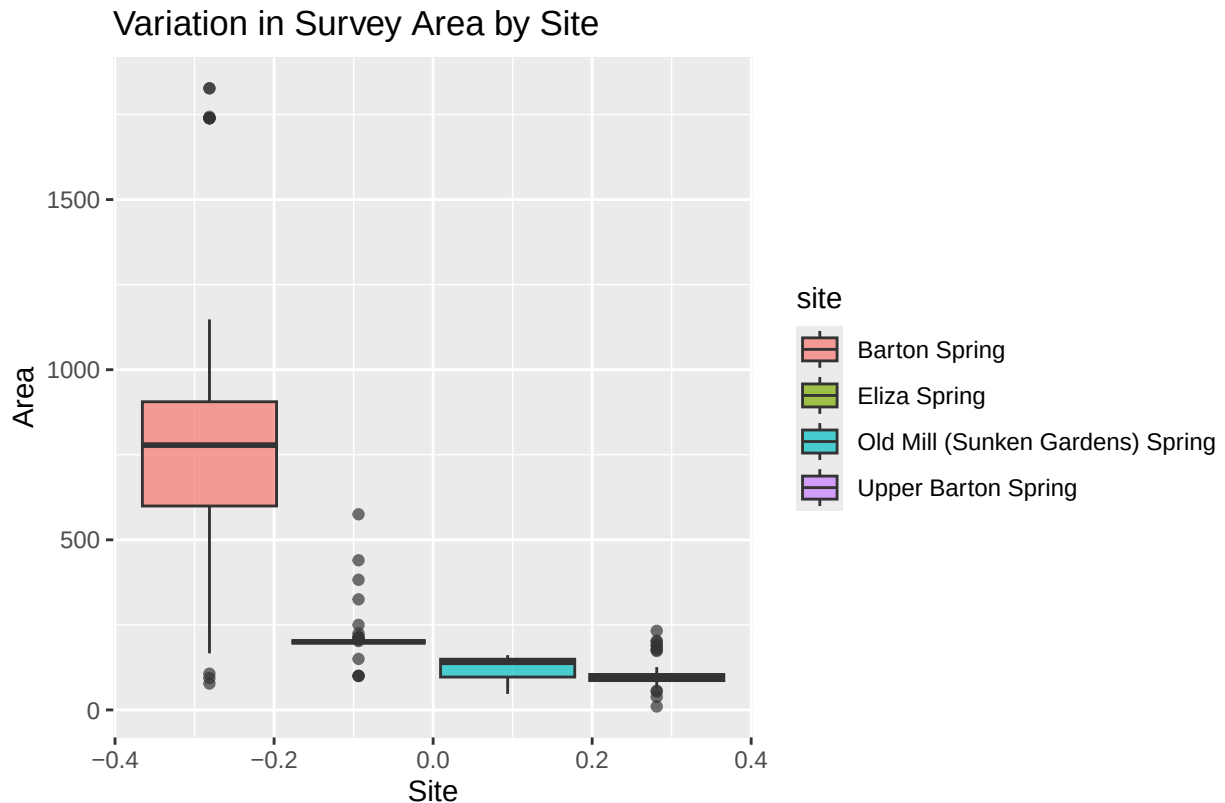
Variation in Survey Area by Site



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_boxplot(aes(y = area, fill = site), alpha = 0.7) +
  labs(title = "Variation in Survey Area by Site",
        x = "Site",
        y = "Area",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2152 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



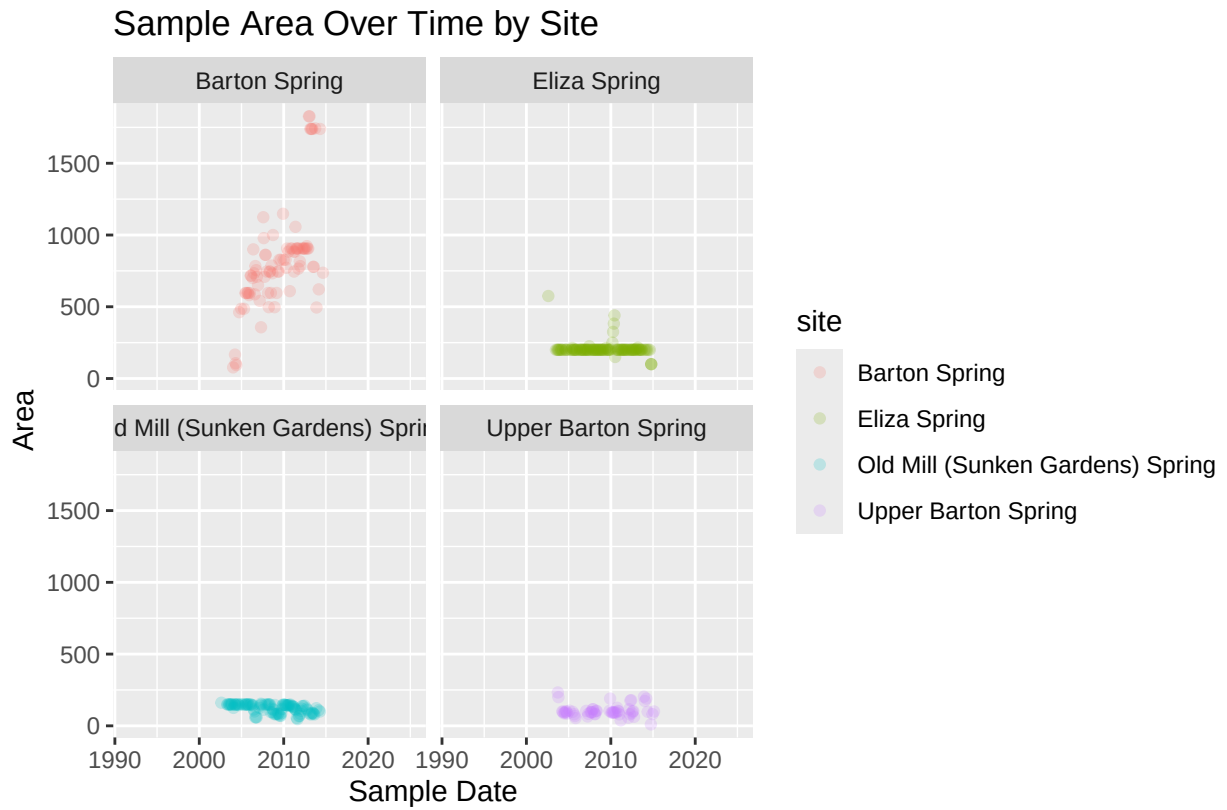
Data from the City of Austin Barton Springs Salamanders DO and Flow

While the areas sampled for Eliza, Old Mill (Sunken Garden), and Upper Barton Springs were relatively consistent, Barton Spring's sample areas were much more spread, and also had the greatest outlier among them all. This raises an important question to keep in mind going forward: if Barton Spring is sampled over a much larger and more variable area than the other sites, any observed differences in DO or salamander detection at Barton Spring may partly reflect differences in survey effort rather than the site itself

We were then interested in the when theses surveys took place, and if anything could be noticed to that helps explain the Barton Springs outliers

```
water_cleaned |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = area, color = site), alpha = 0.2) +
  facet_wrap(~site) +
  labs(title = "Sample Area Over Time by Site",
        x = "Sample Date",
        y = "Area",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

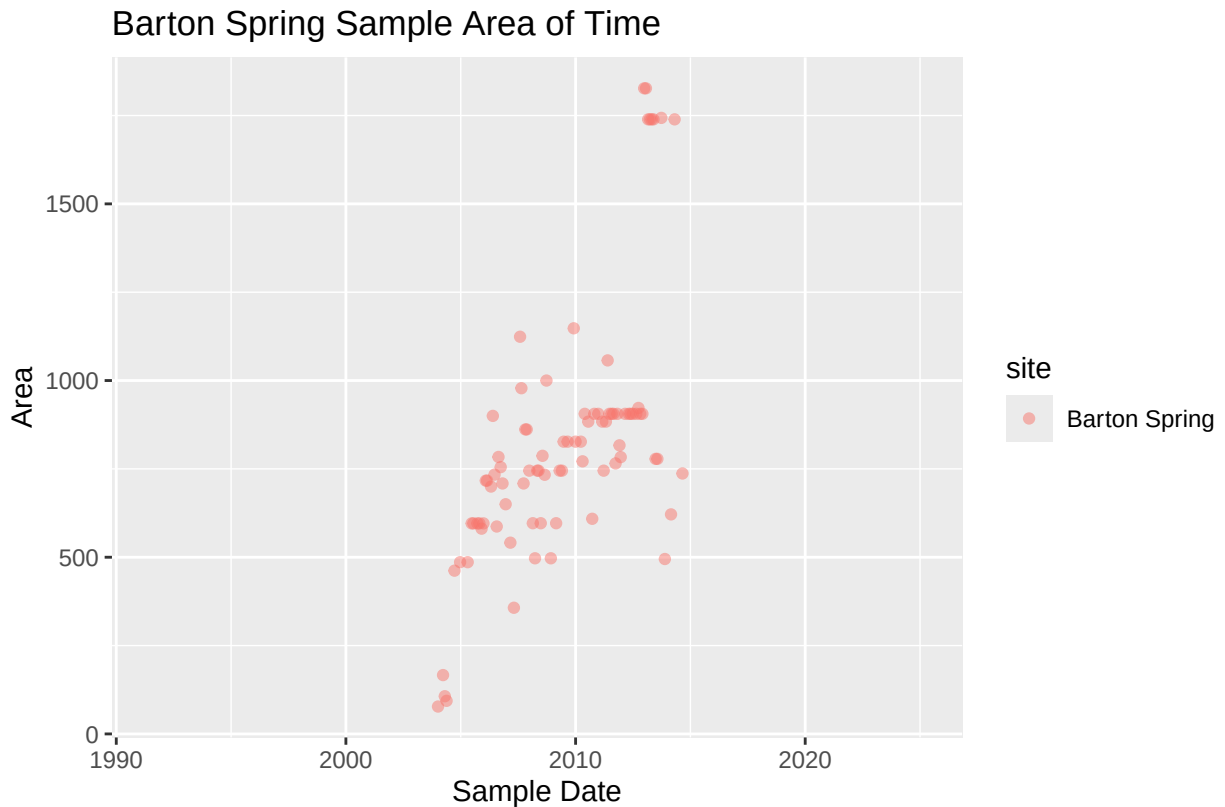
```
## Warning: Removed 2152 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
water_cleaned |>
  filter(site == "Barton Spring") |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = area, color = site), alpha = 0.5) +
  labs(title = "Barton Spring Sample Area of Time",
        x = "Sample Date",
        y = "Area",
        caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 878 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



The Barton Spring outliers are not randomly distributed in time. All three occur within a narrow window around 2013, distinctly separated from the site's typical survey area range (roughly 100-1150 sq ft) seen across the rest of the 2004-2015 sampling period. This suggests a specific, likely temporary change in survey protocol or scope at that site around 2013, rather than ongoing variability. Without further documentation from the data source, the exact cause of this change can't be determined here. It is worth flagging as an anomaly isolated to a specific period rather than a general characteristic of Barton Spring sampling.

Taking a closer look at the 3 highest area surveys for Barton Spring

```
# Check high values of area
water_cleaned |>
  slice_max(n = 3, area)
```

```
## # A tibble: 3 x 6
##   sampledte site      DO Flowcode salamanders area
##   <date>      <chr>    <dbl>    <dbl>      <dbl> <dbl>
## 1 2013-01-24 Barton Spring 5.05      NA        NA 1827
## 2 2012-12-27 Barton Spring 5.1       NA        NA 1827
## 3 2013-09-26 Barton Spring 4.8       NA        NA 1743.
```

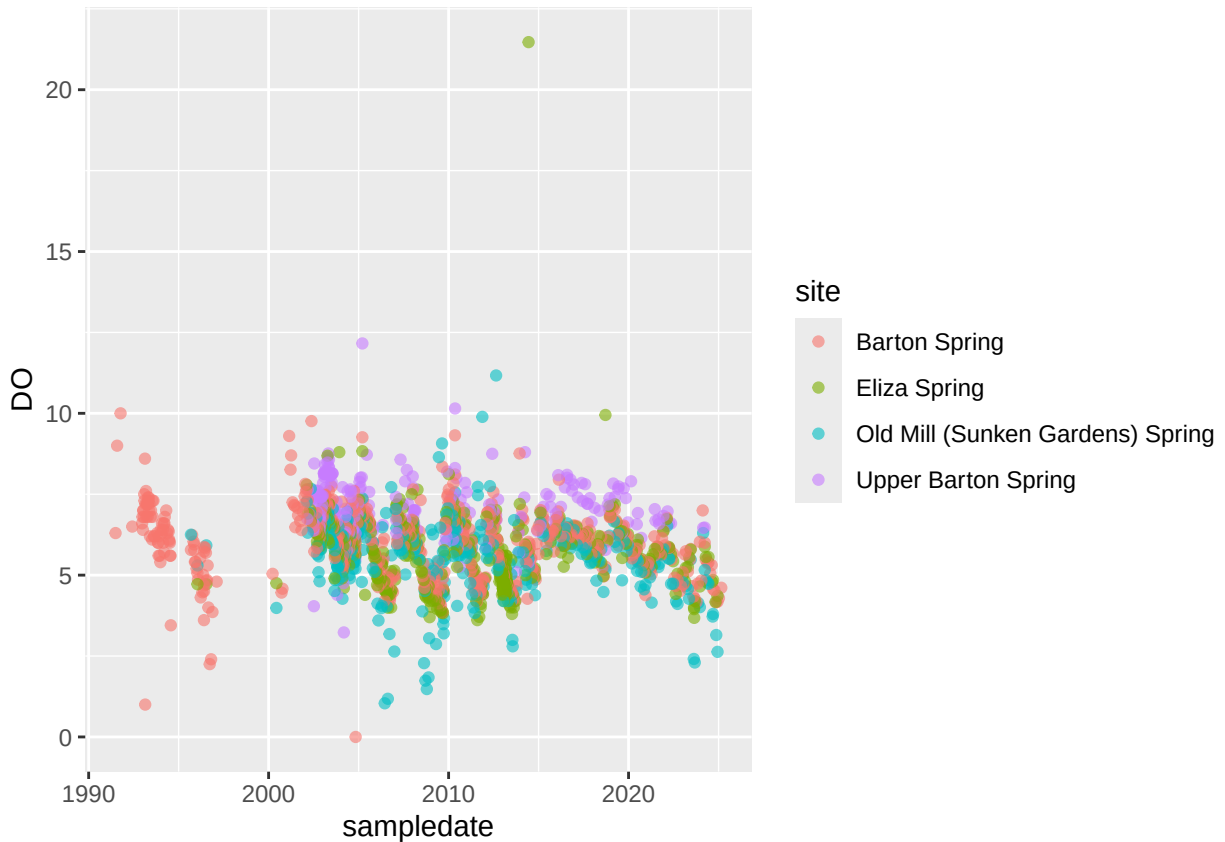
The highest survey area for Barton Spring is approximately 1827 Sq Ft, taking place on two occasions in late 2012/early 2013, with a third closely comparable outlier (1743 Sq Ft) later that year.

With the exception of Barton Spring, most of the sites have consistent survey areas. How do their dissolved oxygen levels vary over time?

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
```

```
geom_point(aes(x = sampleddate, y = DO, color = site), alpha = 0.6)
```

```
## Warning: Removed 893 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



Given how unusual this value is, it's worth verifying the units are correctly recorded and that this isn't an artifact of the sub-sample averaging step, rather than a genuine reading.

```
water |>
  filter(PARAMETER == "DISSOLVED OXYGEN",
         SITE_NAME == "Eliza Spring",
         str_detect(SAMPLE_DATE, "2014")) |>
  select(SAMPLE_DATE, SITE_NAME, RESULT, UNIT, FILTER)
```

```
## # A tibble: 12 x 5
##   SAMPLE_DATE      SITE_NAME  RESULT UNIT  FILTER
##   <chr>           <chr>    <dbl> <chr> <chr>
## 1 10/22/2014 12:35:00 PM Eliza Spring  4.93 MG/L Dissolved
## 2 10/15/2014 09:58:00 AM Eliza Spring  5.21 MG/L Dissolved
## 3 10/14/2014 10:49:00 AM Eliza Spring  5.14 MG/L Dissolved
## 4 08/14/2014 09:30:00 AM Eliza Spring  5.24 MG/L Dissolved
## 5 08/14/2014 09:30:00 AM Eliza Spring  4.6 MG/L Dissolved
## 6 06/12/2014 09:44:00 AM Eliza Spring 21.6 MG/L Dissolved
## 7 06/12/2014 09:44:00 AM Eliza Spring 21.3 MG/L Dissolved
## 8 06/11/2014 11:35:00 AM Eliza Spring  5.7 MG/L Dissolved
## 9 04/24/2014 08:15:00 AM Eliza Spring  5.86 MG/L Dissolved
## 10 04/10/2014 09:30:00 AM Eliza Spring  5 MG/L Dissolved
## 11 04/01/2014 01:10:00 PM Eliza Spring  5.95 MG/L Dissolved
```

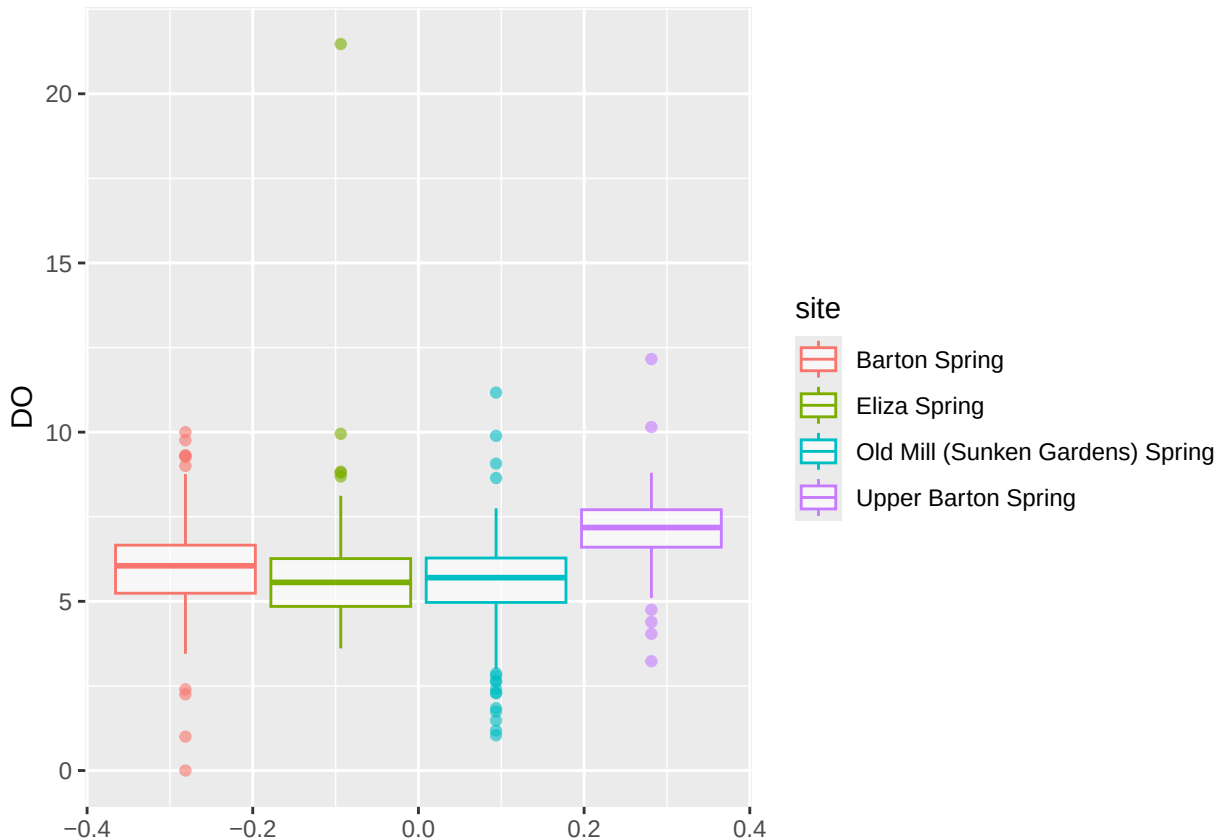
```
## 12 02/13/2014 10:50:00 AM Eliza Spring 6.4 MG/L Dissolved
```

One elevated DO reading at Eliza Spring in June 2014 (~21.5 mg/L, confirmed across two sub-samples on the same visit) stands out from the otherwise consistent 0-13 mg/L range seen throughout the dataset. Unlike the percent-saturation contamination addressed above, this value is correctly recorded in mg/L and is retained as a genuine observation, possibly reflecting a short-term supersaturation event (e.g. algal activity or aeration) rather than a data quality issue. This would be an excellent question for city officials who conducted the survey.

DO levels appear broadly consistent over time across all sites, with the single exception of the Eliza Spring outlier noted above. This raises the question of how DO varies by site specifically.

```
# log10(DO) by site
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_boxplot(aes(y = DO, color = site), alpha = 0.6)
```

```
## Warning: Removed 893 rows containing non-finite outside the scale range
## (`stat_boxplot()`.)
```



DO levels are broadly similar in typical range across the four sites (median values cluster between roughly 5.5 and 7.5 mg/L), but with some notable site-specific patterns. Upper Barton Spring shows the highest typical DO of the four sites. Old Mill (Sunken Gardens) Spring has the tightest core distribution but includes several low-DO outliers, pulling down its overall range. Barton Spring shows the widest spread, including a few near-zero readings. Eliza Spring's single elevated outlier (~21.5 mg/L, discussed above) stands apart from an otherwise typical distribution.

```
water_cleaned |>
  summarize("Total salamander count" = sum(salamanders, na.rm = T))
```

0.3.0.2 Salamanders

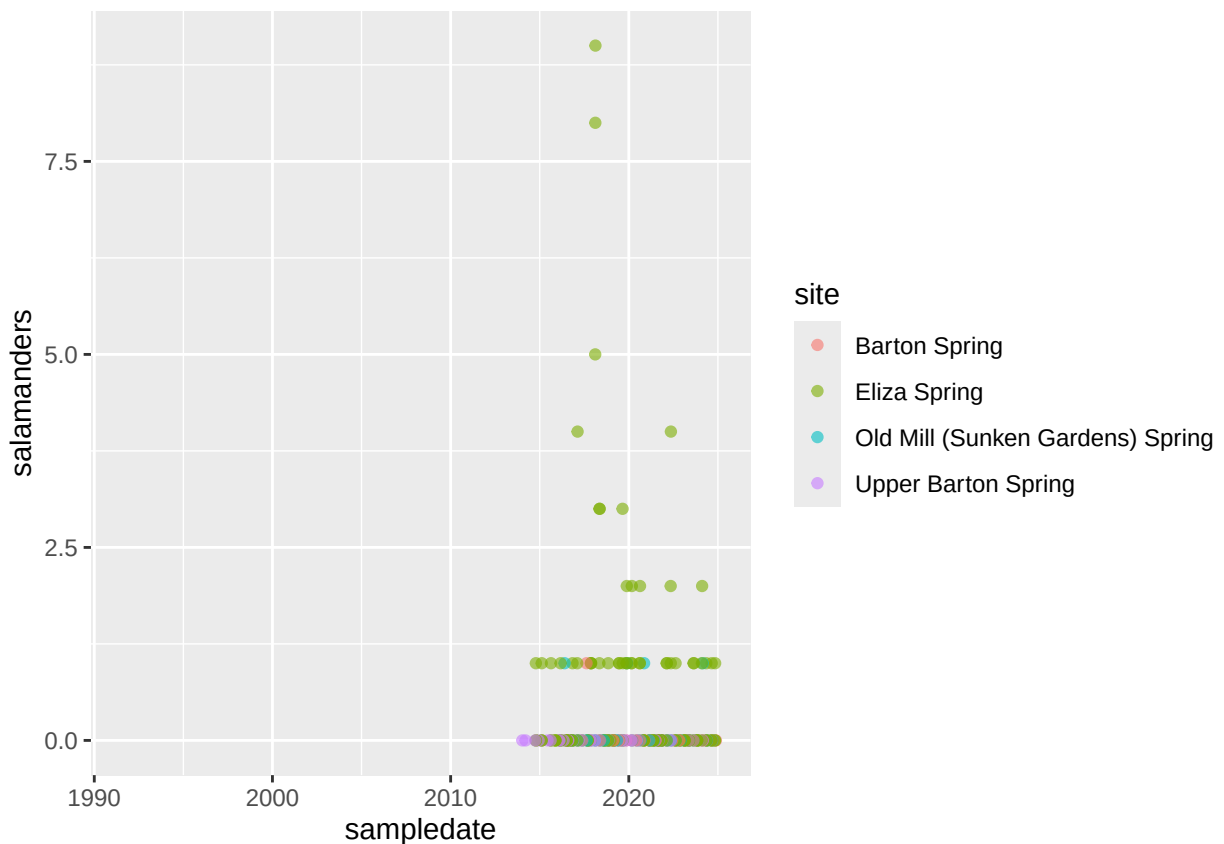
```
## # A tibble: 1 x 1
##   `Total salamander count`
##               <dbl>
## 1                   84
```

A total of 84 salamanders were counted across the entire dataset. This is not to say that there exist 84 salamanders in total though, as several are likely repeated counts over time.

How is this distributed by region?

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = salamanders, color = site), alpha = 0.6)
```

```
## Warning: Removed 2297 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



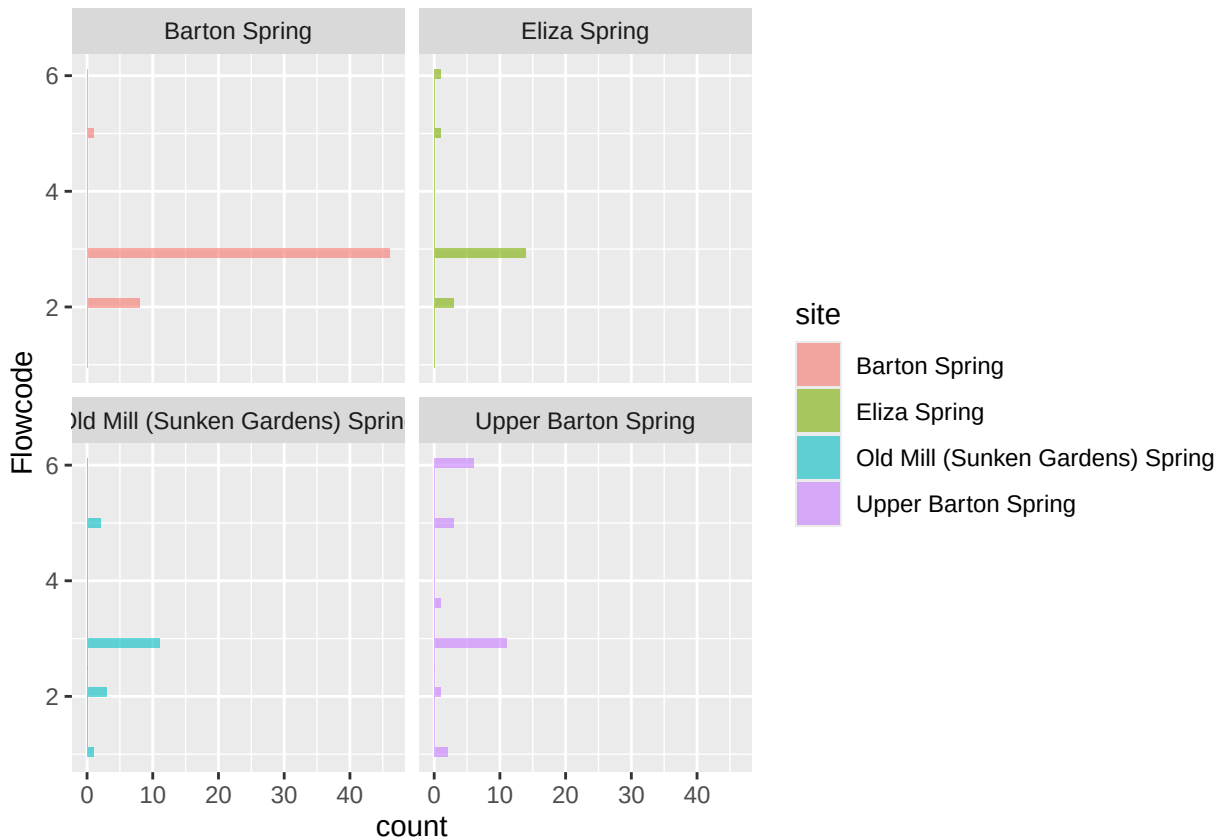
Seems like the salamander counting for this view does not start until around 2015, or that there were no salamanders to count until then, and that they like Eliza Spring to most. Regardless, Eliza Spring's markedly higher salamander detection rate, combined with Barton Spring's earlier-noted variability in survey area, suggests that site-level differences in either measured DO or detected salamanders may partly reflect differences in sampling practices rather than

the underlying spring conditions themselves.

Given this is the case, Does flow perhaps effect DO?

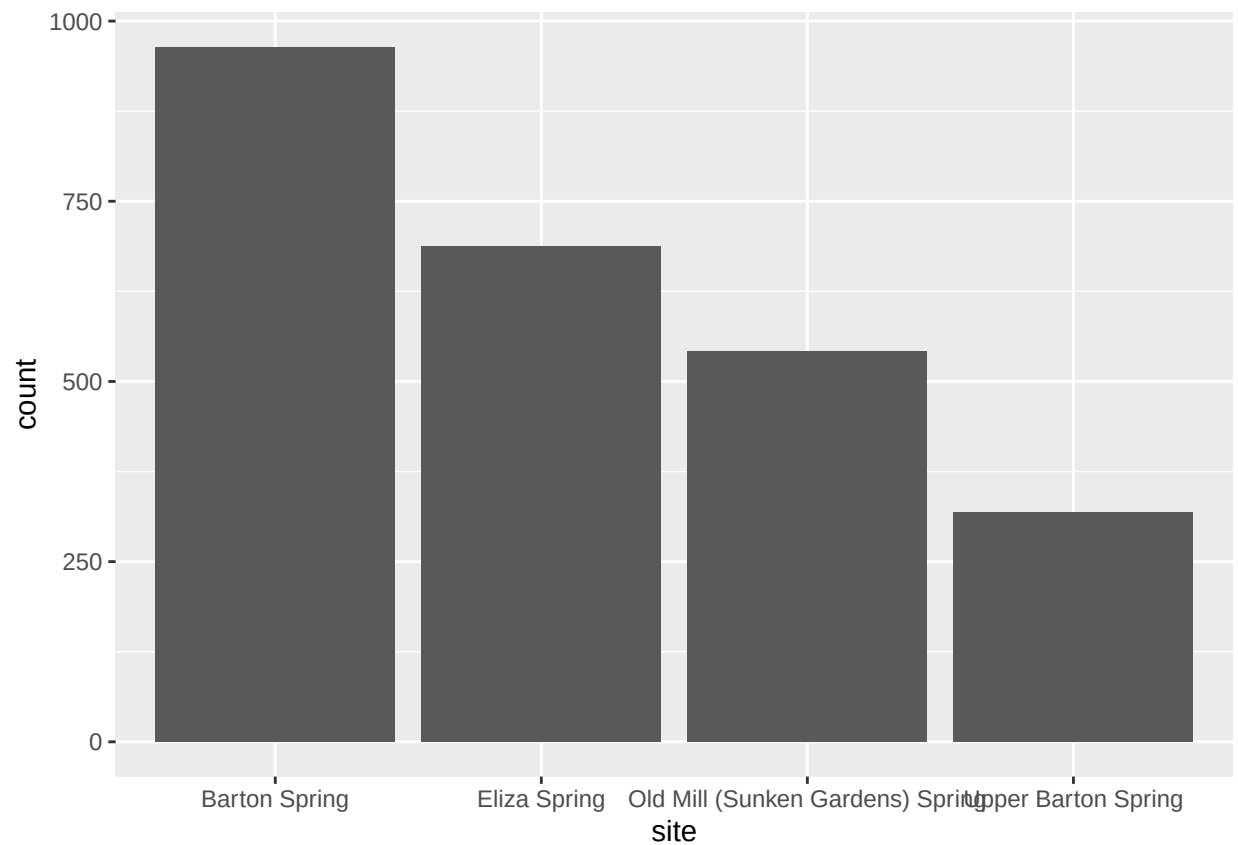
```
water_cleaned |>
  ggplot() +
  geom_histogram(aes(y = Flowcode, fill = site), alpha = 0.6) +
  facet_wrap(~site)
```

```
## Warning: Removed 2397 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



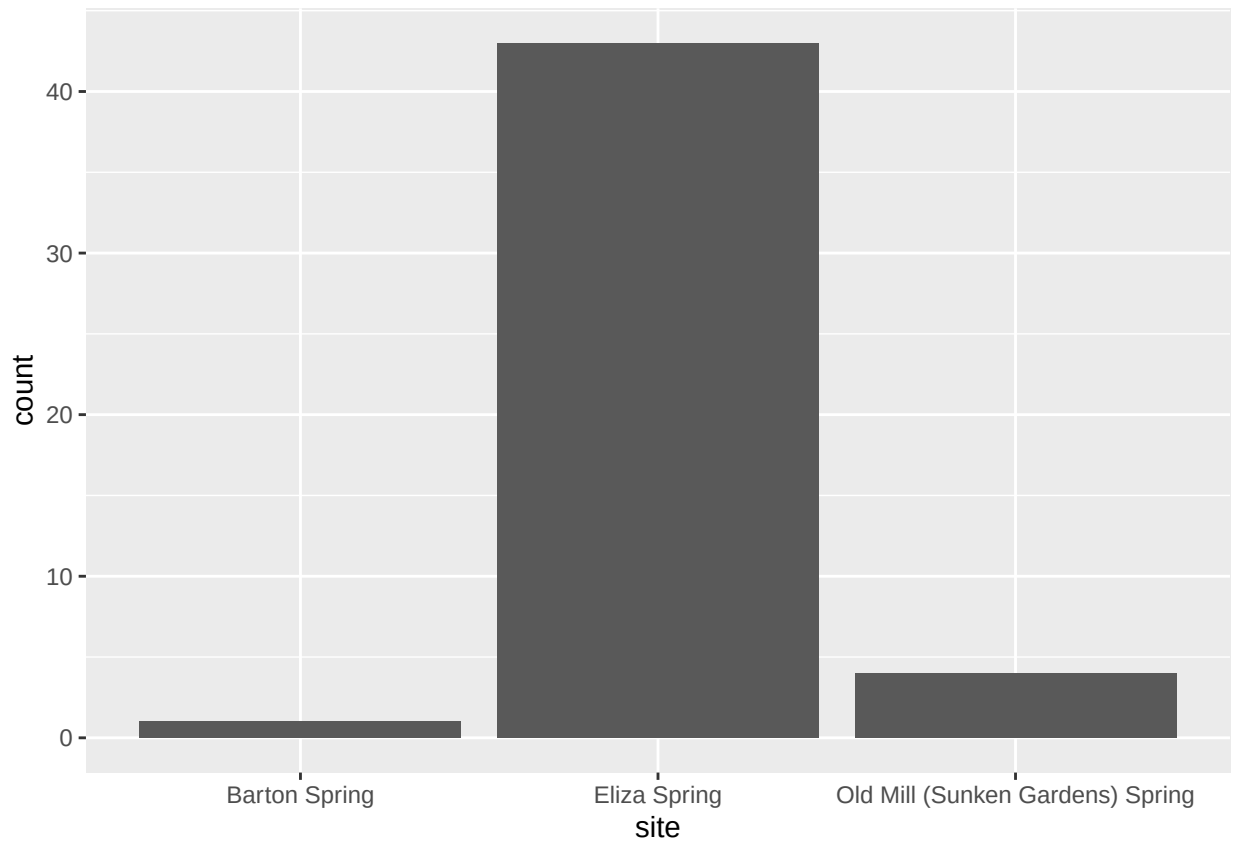
This suggests that Barton Spring and Eliza Spring are somewhat over-represented in the dataset. Let's check for sampling country for each

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_bar(aes(x = site))
```



And wouldn't you believe it, they are indeed the most sampled sites from this view. Lets check where salamanders are being looked for the most

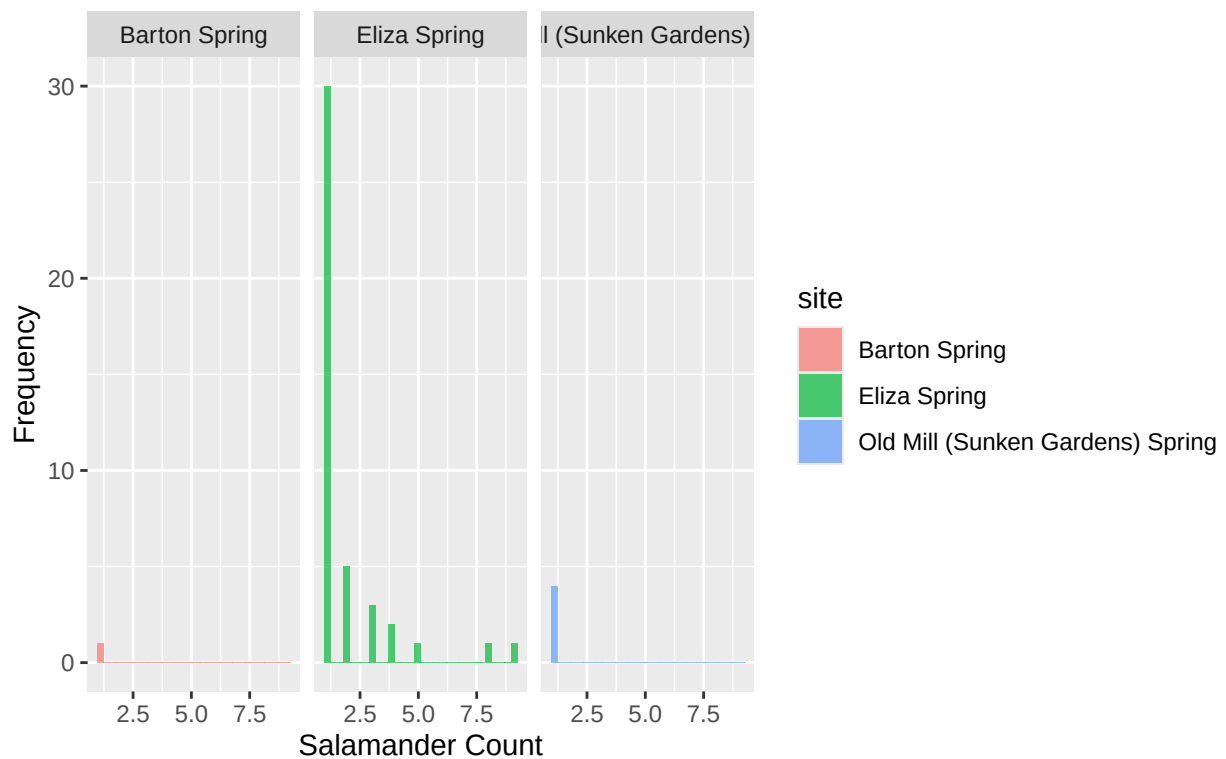
```
water_cleaned |>
  group_by(site) |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_bar(aes(x = site))
```



Salamander detections were recorded at three of the four sites. Eliza Spring far more frequently than the others (over 40 site-visits with at least one detection), Old Mill less often, and Barton Spring with a single incidental detection. Upper Barton Spring had no recorded detections at all

```
water_cleaned |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_histogram(aes(x = salamanders, fill = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Salamander Count Distributions by Spring",
       x = "Salamander Count",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

Salamander Count Distributions by Spring



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_point(aes(x = sampleddate, y = Flowcode, color = site)) +
  facet_wrap(~site) +
  labs(title = "Flowcode Distributions by Spring",
       x = "Flowcode",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2397 rows containing missing values or values outside the scale range
## (`geom_point()`).
```

Flowcode Distributions by Spring



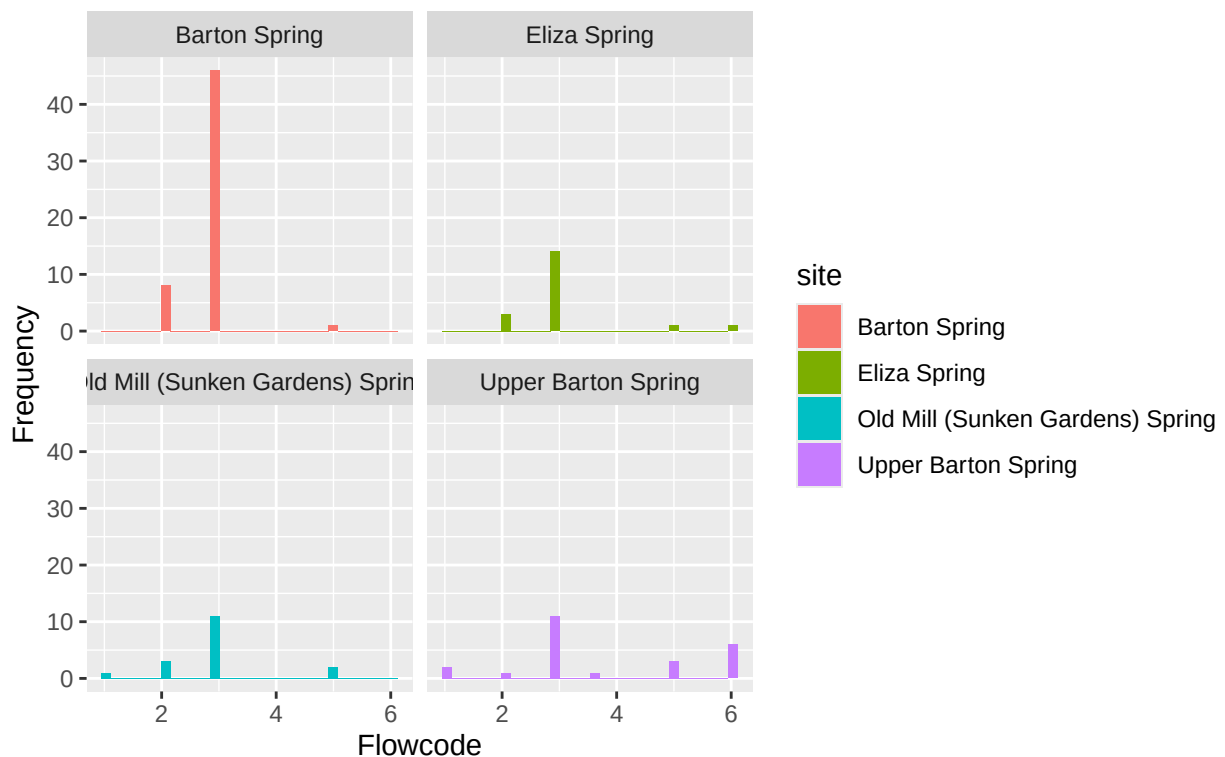
Data from the City of Austin Barton Springs Salamanders DO and Flow

typical flowcode distributions for each

```
water_cleaned |>
  group_by(site) |>
  ggplot() +
  geom_histogram(aes(x = Flowcode, fill = site)) +
  facet_wrap(~site) +
  labs(title = "Flowcode Distributions by Spring",
       x = "Flowcode",
       y = "Frequency",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")
```

```
## Warning: Removed 2397 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

Flowcode Distributions by Spring



Data from the City of Austin Barton Springs Salamanders DO and Flow

Nick's Model

```
# Getting rid of NA values
model_water <- water_cleaned |>
  filter(!is.na(DO))
# Fit model
lin_model <- lm(DO ~ sampleddate + site, data = model_water)
summary(lin_model)
```

```
##
## Call:
## lm(formula = DO ~ sampleddate + site, data = model_water)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.0723 -0.6338  0.0705  0.6427 15.9484
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    7.105e+00  1.548e-01  45.901 < 2e-16 ***
## sampleddate   -8.115e-05  1.056e-05  -7.685 2.64e-14 ***
## siteEliza Spring -2.659e-01  6.837e-02  -3.890 0.000104 ***
## siteOld Mill (Sunken Gardens) Spring -3.589e-01  7.605e-02  -4.719 2.58e-06 ***
## siteUpper Barton Spring  1.207e+00  8.657e-02  13.944 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.099 on 1614 degrees of freedom
```

```

## Multiple R-squared:  0.2008, Adjusted R-squared:  0.1989
## F-statistic: 101.4 on 4 and 1614 DF,  p-value: < 2.2e-16

# Calculating RMSE
sqrt(mean(resid(lin_model)^2))

## [1] 1.097667

# Number of folds
k = 5

# Setting a seed
set.seed(322)

# Randomly order rows in the dataset
shuffled_data <- model_water[sample(nrow(model_water)),]

# Create k folds in the dataset
shuffled_data <- shuffled_data |>
  mutate(folds = cut(seq(1:nrow(shuffled_data)), breaks = k, labels = FALSE))

# Initialize a vector to keep track of the performance for each k-fold
perf_k <- NULL

# Use a for-loop to get performance for each k-fold
for(i in 1:k){
  # Split data into train and test data
  train_i <- shuffled_data |> filter(folds != i) #train data = all observations except in fold i
  test_i <- shuffled_data |> filter(folds == i) # test data = observations in fold i

  # Train model on train data (all but fold i)
  train_model <- lm(DO ~ site + sampleddate,
                    data = train_i)

  # Performance listed for each test data = fold i
  sal_per <- ggplot(test_i) +
    geom_roc(aes(
      # Outcome is DO
      d = DO,
      # Probability of Pokemon legendary status on logistic model
      m = predict(train_model, newdata = test_i)[ ,2]))
  # Save the AUC values
  perf_k[i] <- sqrt(mean(resid(train_model)^2))
}

# Performance for each fold
perf_k

## [1] 1.142751 1.120624 1.108006 1.021135 1.089821

# Average performance over all k folds
mean(perf_k)

## [1] 1.096467

My Model

```

```

# Getting rid of NA values
model_water2 <- water_cleaned |>
  filter(!is.na(salamanders))
# Fit model
lin_model2 <- lm(salamanders ~ sampleddate + site, data = model_water2)
summary(lin_model2)

##
## Call:
## lm(formula = salamanders ~ sampleddate + site, data = model_water2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8158 -0.6158 -0.0722  0.0786  8.2670
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      1.284e+00  1.307e+00   0.982  0.32724
## sampleddate      -6.787e-05  7.004e-05  -0.969  0.33367
## siteEliza Spring    6.423e-01  2.019e-01   3.181  0.00169 **
## siteOld Mill (Sunken Gardens) Spring  7.962e-02  2.493e-01   0.319  0.74975
## siteUpper Barton Spring -8.740e-02  2.664e-01  -0.328  0.74318
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.042 on 210 degrees of freedom
## Multiple R-squared:  0.0914, Adjusted R-squared:  0.07409
## F-statistic: 5.281 on 4 and 210 DF,  p-value: 0.0004513

# Calculating RMSE
sqrt(mean(resid(lin_model2)^2))

## [1] 1.030295

# Number of folds
k = 5

# Setting a seed
set.seed(322)

# Randomly order rows in the dataset
shuffled_data <- model_water2[sample(nrow(model_water2)),]

# Create k folds in the dataset
shuffled_data <- shuffled_data |>
  mutate(folds = cut(seq(1:nrow(shuffled_data)), breaks = k, labels = FALSE))

# Initialize a vector to keep track of the performance for each k-fold
perf_k <- NULL

# Use a for-loop to get performance for each k-fold
for(i in 1:k){
  # Split data into train and test data
  train_i <- shuffled_data |> filter(folds != i) #train data = all observations except in fold i
  test_i <- shuffled_data |> filter(folds == i) # test data = observations in fold i

```

```

# Train model on train data (all but fold i)
train_model <- lm(salamanders ~ site + sampleddate,
                  data = train_i)

# Performance listed for each test data = fold i
sal_per <- ggplot(test_i) +
  geom_roc(aes(
    # Outcome is salamanders
    d = salamanders,
    # Probability of Pokemon legendary status on logistic model
    m = predict(train_model, newdata = test_i)[,2]))
# Save the AUC values
perf_k[i] <- sqrt(mean(resid(train_model)^2))
}

# Performance for each fold
perf_k

## [1] 1.1112771 1.0784370 1.1201961 0.6174242 1.1115565

# Average performance over all k folds
mean(perf_k)

## [1] 1.007778

water_cleaned |>
  filter(salamanders != 0) |>
  ggplot() +
  geom_point(aes(y = salamanders, x = sampleddate, color = site), alpha = 0.7) +
  facet_wrap(~site) +
  labs(title = "Salamander Measurements by Spring",
       x = "Time",
       y = "Salamander Count",
       caption = "Data from the City of Austin Barton Springs Salamanders DO and Flow")

```

Salamander Measurements by Spring



Data from the City of Austin Barton Springs Salamanders DO and Flow

```
summary(water_cleaned$salamanders)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 0.0000  0.0000  0.0000  0.3907  0.0000  9.0000    2297
```

```
water_cleaned |>
  filter(!is.na(salamanders)) |>
  count(n())
```

```
## # A tibble: 1 x 2
##   `n()`      n
##   <int> <int>
## 1    215    215
```

0.4 Discussion

0.5 Reflection, Acknowledgements, & References. Measurement

Huge thank you to Mr. Robert Clayton, who's help was integral to this work